MACHINE LEARNING AND APPLICATIONS

Literature Survey, Research Gaps and Proposed Features

Bank Term-Deposit Subscription Prediction Using Machine Learning

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PROJECT FOUNDATION

Scope of the Proposed Work

A deliberately manageable finance-domain classification project with a clear operational outcome and no requirement for trading or market-microstructure expertise.

Problem statement. Given a bank customer's demographic, financial and previous-campaign information, predict the probability that the customer will subscribe to a term deposit. The operational goal is to prioritize a limited contact list while avoiding unnecessary calls.

DATASET	ML TASK	FINAL OUTPUT
UCI Bank Marketing: **45,211 records**, **16 predictors**, mixed numerical/categorical fields and a binary yes/no target.	Binary classification with Logistic Regression, Decision Tree and Random Forest; XGBoost is optional, not compulsory.	A probability, contact recommendation, explanation and batch-ranked customer list delivered through a Streamlit application.

Important design decision	How it changes this project
Current-call duration is known only after the call ends.	The deployable **pre-call model excludes duration**. A second post-call model may include it only for analysis. The performance difference becomes an explicit leakage/availability audit.
Most papers report aggregate predictive performance.	This project adds calibrated probabilities, per-customer explanations and an adjustable decision threshold so that model output becomes an operational recommendation.
The public data comes from one Portuguese bank and an older campaign period.	The report does not claim universal performance. The limitation is documented in the model card, and chronological validation is used where record ordering is available.

Reading the gap table. The limitations on the following pages are gaps relative to this proposed CBDP, derived from the reported scope of each study. They are not presented as errors or as limitations necessarily claimed by the original authors.

SECTION 1

Related Research Papers (1-6 of 12)

Chronological coverage of benchmark models, feature engineering, deep learning, class-imbalance methods, ensembles and explainable approaches.

ID	Paper	Venue	Data / Method	Main Contribution
P1	**Moro, Cortez and Rita (2014)** A data-driven approach to predict the success of bank telemarketing	Decision Support Systems, 62, 22-31	Portuguese retail-bank campaigns (2008-2013); 150 candidate variables reduced to 22; Logistic Regression, Decision Tree, Neural Network and SVM.	Established the benchmark problem and data-mining pipeline. The Neural Network produced the strongest AUC, while sensitivity analysis identified influential inputs.
P2	**Moro, Cortez and Rita (2015)** Using customer lifetime value and neural networks to improve the prediction of bank deposit subscription	Neural Computing and Applications, 26, 131-139	Twelve customer-lifetime-value/history variables; forward selection; multilayer perceptron; realistic rolling-window evaluation.	Validated five history-based features and improved cumulative lift by up to four percentage points over a no-history baseline.
P3	**Kim, Lee, Jo and Cho (2016)** Predicting the success of bank telemarketing using deep convolutional neural network	IEEE SoCPaR, pp. 314-317	Deep convolutional neural network applied to 45,211 bank-marketing calls; architecture and learning parameters were explored.	Demonstrated a deep-learning alternative and reported 76.70% accuracy against conventional classifiers.
P4	**Moro, Cortez and Rita (2018)** A divide-and-conquer strategy using feature relevance and expert knowledge	Expert Systems, 35(4), e12253	Sensitivity analysis plus expert evaluation; call direction (inbound/outbound) used to divide the telemarketing problem into sub-problems.	Showed that sub-problem modeling can materially improve inbound-call targeting performance.
P5	**Jiang (2018)** Using Logistic Regression Model to Predict the Success of Bank Telemarketing	International Journal on Data Science and Technology, 4(1), 35-41	Logistic Regression compared with Bayes, SVM, Neural Network and Decision Tree; experiments implemented in R.	Provided a transparent statistical baseline and reported the strongest accuracy and ROC-AUC for Logistic Regression in its experiments.
P6	**Ghatasheh et al. (2020)** Business Analytics in Telemarketing: Cost-Sensitive Analysis of Bank Campaigns Using ANNs	Applied Sciences, 10(7), 2581	Cost-sensitive multilayer perceptron and MetaCost strategy for imbalanced campaign data, without resampling the original observations.	Focused on unequal error costs; reported 79% G-mean and Type I/II errors of 0.192/0.229 for the best model.

Full citations and DOI links appear in the References section.

SECTION 1

Related Research Papers (7-12 of 12)

Chronological coverage of benchmark models, feature engineering, deep learning, class-imbalance methods, ensembles and explainable approaches.

ID	Paper	Venue	Data / Method	Main Contribution
P7	**Yan, Li and Liu (2020)** Prediction of bank telephone marketing results based on improved whale algorithms optimizing S_Kohonen network	Applied Soft Computing, 92, 106259	An improved whale-optimization algorithm tunes S_Kohonen network weights using nonlinear adjustment and Gaussian perturbation.	Improved search and network stability compared with the unoptimized approach and benchmark optimizers.
P8	**Feng, Yin, Wang and Dhamotharan (2022)** A dynamic ensemble selection method for bank telemarketing sales prediction	Journal of Business Research, 139, 368-382	META-DES-AAP dynamically selects base classifiers through multi-objective optimization of accuracy and average profit.	Connected model selection to campaign profit and reported leading accuracy and average profit against comparison methods.
P9	**Tékouabou et al. (2022)** A Machine Learning Framework towards Bank Telemarketing Prediction	Journal of Risk and Financial Management, 15(6), 269	Class Membership-Based (CMB) treatment of categorical/dummy variables and a prediction framework evaluated on bank-campaign data.	Reported 97.3% accuracy and 95.9% AUC, emphasizing more principled categorical encoding and stable aggregate performance.
P10	**Xie et al. (2023)** How to improve the success of bank telemarketing? Prediction and interpretability analysis	Computers and Industrial Engineering, 175, 108874	Random Subspace, Multi-Boosting and RS-MB ensembles; variable selection; partial-dependence plots for global effects.	Combined strong ensemble prediction with interpretation; RS-MB using selected variables performed best in the reported experiments.
P11	**Ghatasheh, Altaharwa and Aldebei (2023)** Modeling the Telemarketing Process Using Genetic Algorithms and Extreme Boosting	IEEE Access, 11, 67806-67824	Genetic algorithm simultaneously selects features and tunes cost-sensitive XGBoost; 50-times repeated stratified 10-fold cross-validation.	Achieved 89.07% G-mean and 0.059 Type I error while using a small feature subset and preserving the original class distribution.
P12	**Peter et al. (2025)** Predicting customer subscription in bank telemarketing campaigns using ensemble learning models	Machine Learning with Applications, 19, 100618	Bagging, boosting and stacking ensembles with feature-importance analysis on the bank-marketing data.	Stacking achieved 91.88% accuracy and 0.9491 ROC-AUC; duration, Euribor and age were identified as influential.

Full citations and DOI links appear in the References section.

SECTION 2

Research Gaps Relative to Our Project (1-6 of 12)

Each row links the paper's reported focus to a specific project response. This avoids claiming novelty that is not actually implemented.

ID	Paper Focus	Gap / Limitation Relevant to the CBDP	How Our Project Responds
P1	Benchmark prediction, feature selection and model comparison.	Evidence comes from one bank and period. The study does not provide a separate deployable pre-call model versus a post-call analysis model, nor calibrated probabilities, fairness checks or a user application.	F1 pre-call model; F2 post-call comparison; F3 chronological testing; F5 calibration; F8 fairness; F9 application.
P2	Customer history and lifetime-value features.	The richer history variables may not exist in the standard public dataset, reducing reproducibility for a course project. The neural model emphasizes lift rather than simple per-customer explanations and deployment controls.	Use reproducible UCI fields; engineer transparent history indicators (E1-E4); add local explanations and a lightweight app.
P3	Deep learning on structured campaign data.	A CNN increases implementation and explanation complexity for tabular data. Evaluation is centered on accuracy, with limited treatment of minority-class utility, business costs, probability calibration and explanations.	Prefer interpretable tabular models; report PR-AUC/F1/recall; add class weights, calibration, cost-aware thresholds and SHAP.
P4	Expert-guided segmentation and feature relevance.	The split depends on call-direction information and expert decisions that may not transfer to every dataset. Probability calibration, customer-level explanations and deployable workflow are outside its main scope.	Use only available UCI fields; make preprocessing reproducible; add calibrated probabilities, local explanations and Streamlit delivery.
P5	Simple, interpretable classification baseline.	The work does not center class-imbalance handling, cost-sensitive threshold selection, chronological validation or individual prediction explanations.	Retain Logistic Regression as baseline; add class weighting, time-aware split, calibrated thresholds, PR-AUC and SHAP/local factors.
P6	Class imbalance and unequal decision costs.	The ANN remains comparatively difficult to explain. The reported scope does not combine chronological testing, pre-call feature availability checks, calibrated outputs, fairness or a simple user-facing tool.	Use class-weighted interpretable models; test chronologically; separate pre/post-call inputs; calibrate and explain each prediction.

Common cross-paper gap: strong offline scores are often emphasized, but a deployable decision also needs feature availability at scoring time, calibrated probabilities, an action threshold and transparent reasons.

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Research Gaps Relative to Our Project (7-12 of 12)

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ID	Paper Focus	Gap / Limitation Relevant to the CBDP	How Our Project Responds
P7	Metaheuristic optimization of a neural classifier.	The hybrid optimizer is difficult to reproduce, explain and maintain in a small CBDP. Calibration, fairness, operational cost thresholds and end-user deployment are not the central contribution.	Use standard scikit-learn models with fixed seeds and saved pipelines; focus novelty on deployment, decisions and explanations.
P8	Dynamic ensembles and profit-aware learning.	Results depend on specified revenue and cost assumptions, while the ensemble is complex for a first implementation. Local explanations, fairness and a pre-call/post-call distinction remain open integration needs.	Use a transparent cost matrix and adjustable threshold; retain simple models; expose local factors and separate deployable inputs.
P9	Categorical representation and aggregate performance.	The main contribution is encoding and model performance on one campaign setting. It does not integrate cost-aware thresholding, individual explanations, fairness, chronological validation and a deployable interface.	Implement leakage-safe preprocessing pipeline; add F3-F9 evaluation, decision and deployment features.
P10	Ensemble prediction and global interpretability.	Partial-dependence plots explain overall patterns but not why one specific customer received a score. Calibration, action thresholds and the pre-call availability of variables are not the primary focus.	Add per-customer SHAP explanations, probability calibration, contact thresholding and a strict pre-call feature set.
P11	Cost-sensitive XGBoost, feature selection and optimization.	Repeated stratified folds may mix observations from different campaign periods, while GA tuning adds computational complexity. Calibration, fairness and a compact operational interface are not combined.	Add an out-of-time holdout; use lighter tuning; calibrate outputs; audit groups; publish a reproducible app and model card.
P12	Recent ensemble comparison and feature importance.	Call duration is highly predictive but unavailable before a call is completed. The reported framework does not explicitly audit this deployment-time issue or combine local explanation, calibration and fairness.	Train and compare pre-call and post-call models; label duration as post-call-only; add local explanations, calibration and group audit.

The proposed novelty is an end-to-end integration novelty for a CBDP. The project does not claim to invent a new classification algorithm.

SECTION 3

New System and Research Features Added by Us

The scope stays implementation-friendly: the novelty comes from combining deployment-aware features, reliable evaluation and an actionable interface.

Priority	New Feature	Implementation	Gap Addressed
Core	F1. Deployable pre-call model	Predict subscription probability using only information available before the current call. Current-call duration is excluded.	Closes the deployment-time feature gap found across much of the literature.
Core	F2. Post-call analysis model and leakage audit	Train a second model that may include call duration, then quantify the performance difference against F1.	Shows why a high offline score may not be usable for pre-call targeting.
Core	F3. Chronological validation	Use older records for training, a later block for validation and the newest block for final testing whenever ordering is available.	Provides a more realistic test than randomly mixing campaign periods.
Core	F4. Imbalance-aware training and metrics	Use class weights first; compare SMOTE only inside training folds. Report recall, precision, F1, PR-AUC, ROC-AUC and confusion matrix.	Prevents majority-class accuracy from hiding missed subscribers.
Core	F5. Calibrated probabilities	Calibrate the best classifier using validation data and verify a reliability curve/Brier score.	Makes a 70% score interpretable as an estimated likelihood, not merely a ranking.
Core	F6. Cost-aware threshold and top-N contact list	Let the bank set call cost and expected deposit benefit, or choose a daily contact capacity; rank customers and choose a transparent threshold.	Converts prediction into an operational decision without a complex dynamic ensemble.
Core	F7. Global and local explainability	Show global importance plus SHAP/local reason codes for each prediction, such as prior success, balance band or contact recency.	Extends global-only interpretation into customer-level explanations.
Recommended	F8. Fairness and responsible-use audit	Compare selection rate, recall and false-positive rate across broad age groups; document that the model supports human review rather than automatic exclusion.	Adds a governance layer rarely combined with telemarketing prediction.
Core	F9. Streamlit application and optional REST API	A form returns probability, recommendation and explanation; a batch page uploads CSV and exports a prioritized contact list.	Turns the notebook into an end-to-end ML application suitable for demonstration.
Recommended	F10. Reproducible model card	Store preprocessing and model in one pipeline, fix random seeds, validate inputs and document data, metrics, limitations and intended use.	Improves reproducibility, safe handoff and evaluation transparency.

SECTION 3A

New Engineered Input Features

All transformations are fitted or defined using training data only. They are deliberately simple enough to explain during a guide review.

Engineered Field	Construction	Purpose	Used In
E1. was_previously_contacted	1 when **pdays != -1**; otherwise 0.	Distinguishes new prospects from customers with prior campaign history.	Pre-call model
E2. previous_success_flag	1 when **poutcome = success**; otherwise 0.	Creates an interpretable signal from the previous campaign outcome.	Pre-call model and explanation
E3. total_contact_count	**campaign + previous**.	Summarizes current and historical contact exposure.	Pre-call model
E4. contact_pressure	**campaign / (previous + 1)**.	Flags heavy current-campaign contact relative to prior history.	Pre-call model and cost analysis
E5. debt_product_count	Count of yes values across housing loan, personal loan and default.	Provides a simple, transparent view of existing credit commitments.	Pre-call model and explanation
E6. balance_segment	Low / medium / high bands derived from training-set quantiles only.	Supports non-linear balance effects without using test information.	Model and dashboard
E7. recency_band	Not contacted, 0-30, 31-90 or more than 90 days using **pdays**.	Makes previous-contact recency easier to explain than a raw value.	Model and dashboard
E8. contact_quarter	Map month to Q1, Q2, Q3 or Q4.	Captures seasonal effects with fewer categories.	Pre-call model
E9. age_group_audit	Less than 30, 30-44, 45-59 and 60+.	Used for fairness reporting and explanations; raw age remains available to the model only after review.	Audit, not automatic exclusion

Minimum viable project	Recommended extension	Keep optional
F1, F2, F4, F6, F7 and F9: pre/post-call models, imbalance metrics, decision threshold, explanations and Streamlit.	F3 and F5: chronological holdout and calibrated probabilities. Add F8 if time permits.	Do not add deep learning, genetic algorithms or real-time banking APIs unless the guide explicitly requires them.

Expected course-project contribution: a reproducible and explainable decision-support system that distinguishes a realistic pre-call prediction from a post-call analytical model, quantifies the effect of unavailable variables, and converts calibrated probabilities into a transparent contact list.

SECTION 3B

Gap-to-Feature Traceability

A compact view of why each proposed addition exists and what evidence will demonstrate it.

Observed Gap	Feature(s)	Evidence Produced by Our Project
Post-call variables can inflate an offline targeting score.	F1, F2	Side-by-side pre-call and post-call metrics, with duration explicitly marked as unavailable before contact.
Random splits may overstate performance across campaign periods.	F3	A final out-of-time test block and a comparison with the random-split result.
Accuracy hides minority-class failures.	F4	Subscriber recall, precision, F1, PR-AUC and confusion matrix, alongside ROC-AUC.
A raw model score is not necessarily a reliable probability.	F5	Calibration curve and Brier score before and after calibration.
Predictions do not directly specify whom to call.	F6	Top-N contact list, adjustable threshold, expected calls saved and conversion yield at selected capacity.
Global feature importance does not explain one customer.	F7	Local reason codes/SHAP values displayed with every single prediction.
Targeting may create uneven errors across customer groups.	F8	Group selection, recall and false-positive-rate table plus responsible-use statement.
Research often stops at a notebook.	F9, F10	Working form, batch upload/export, saved pipeline, input validation and model card.

NOVELTY STATEMENT FOR GUIDE REVIEW

Our work does not claim a new machine-learning algorithm. Its contribution is a **deployment-aware and explainable workflow**: pre-call versus post-call modeling, time-aware validation, calibrated probabilities, cost-aware contact prioritization, local explanations, fairness checks and a working application in one reproducible system.

SOURCES

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Full bibliographic details and persistent links for the literature survey and dataset.

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